# Saamir Fehmi

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## EDUCATION

**University of Ottawa** Sep 2025 – Apr 2030 (Expected)
*Bachelor of Applied Science, Mechanical Engineering — Co-op Program* *Seeking co-op: May 2027*

## ENGINEERING EXPERIENCE

**Suspension Team Member — uOttawa Baja SAE** Sep 2025 – Present
*University of Ottawa Baja SAE Team* *Ottawa, ON*

- Machined upper and lower control arm ball joint housings on a lathe from steel stock, then welded the assemblies to drawing spec to hold correct suspension geometry.
- Fabricated structural frame components with MIG and TIG welding to meet chassis rigidity requirements and SAE rulebook safety standards.
- Contributed to a full vehicle build that competed at Baja SAE New York 2026 and passed all three technical inspections.
- Troubleshot a brake light failure, tracing it to a hydraulic pressure switch whose activation pressure was too high for the brake system.
- Read technical drawings and worked with frame, drivetrain and electrical members from fabrication through final assembly.

## PROJECTS & AWARDS

**XR Chemical Engineering Virtual Lab — 1st Place, uOttawa Engineering Design Day** Nov 2025

- Co-developed a Unity VR recreation of the uOttawa Chemical Engineering Pilot Plant, letting users explore equipment and simulate distillation and heat transfer safely.
- Won 1st place among all first-year engineering teams for innovation, technical execution and real-world applicability.

**Accessible Water Entry Lift — GNG2101 Client Design Project** Sep 2026 – Present

- Designing a manually operated dock-to-lake lift for a young client with limited mobility, lowering her 12+ in. to the water; targets 200 lb capacity, 5-year life, under $200 and winter removal in ≤45 min.
- Pivoted from a hydraulic concept to a simpler lever mechanism with steel tube frame, large rotating bearing and sheet metal shell; benchmarked against $6–12.5K commercial pool lifts and built a 1:10 scale prototype.

**AI Stock Paper-Trading Platform (Personal Project)** 2026

- Building a modular Python platform (FastAPI, Streamlit, SQLAlchemy, Pandas) with a market data engine, technical indicators, a risk engine, backtesting and AI-generated buy/sell/hold recommendations.

## WORK EXPERIENCE

**Salesperson & Drone Operator (Freelance) — TILTOP Roofers** May 2025 – Sep 2025

- Flew aerial roof inspections with a DJI drone under a Transport Canada Advanced licence, capturing high-resolution imagery to assess damage and wear on residential and commercial roofs.
- Generated leads through canvassing and client consultations, feeding the quote pipeline for repair and replacement jobs.

**Customer Consultant (Part-Time) — Fehmi Lights** 2022 – Sep 2025

- Advised residential and commercial clients on lighting solutions and closed sales through consultative selling.

## LEADERSHIP

**Co-Founder, Computer Science Club — Jean Augustine Secondary School** Sep 2024 – Jun 2025

- Founded and led a student CS club; organized a school hackathon, handling logistics and mentoring participants.

**Flight Corporal — 892 Snowy Owl Squadron, Royal Canadian Air Cadets** Sep 2018 – Jun 2020

## SKILLS & CERTIFICATIONS

**CAD & Design:** SolidWorks, AutoCAD, Technical Drawing, GD&T
**Fabrication:** MIG & TIG Welding, Lathe, Milling, CNC Machining, Sheet Metal
**Software:** Python (FastAPI, Pandas, SQLAlchemy), Unity (XR/VR), Microsoft Office
**Certifications:** Transport Canada Advanced Drone Pilot Licence; Private Pilot Licence (in progress)
**Languages:** English, French, Urdu, Hindi